

12x - Notes and Theoretical Background

Theoretical Background:

Remembering the definition we gave to an intake: “The system responsible for manipulating a game piece into the robot perimeter” we will now look at another concept of an intake fulfilling this definition - the “Under The Bumper Intake” or UTBI for short.

Much like a SCBI, the UTBI has no Rotating assembly, but unlike the SCBI, in an UTBI the bumper is very much of a constraint. This creates a problem: how do you insert the game piece into the robot perimeter without rotating the intake and without passing through the bumper? The answer is both counter intuitive and simple - under the bumper (you probably guessed this by now...)

An UTBI is “**An intake in which the path of the game piece goes under the bumper**”, it is usually achieved by maximising the gap between the floor and the bottom side of the bumper while keeping the height delta legal, therefore it can only be used for relatively short pieces that can fit in that gap.

When modeling an UTBI there is almost always a live shaft that is almost touching the floor and another above the game piece in a way that creates compression with the floor and later with the lower shaft. These two shafts are what’s responsible for getting the game piece from the ground and into the robot.



example of an UTBI